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SNAP Linearity

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1. INTRODUCTION

We test the linearity of the SNAP-measured power as a function of input power using a broadband noise source with variable attenuation. Both the ADC sample RMS and the integrated autocorrelation power from the correlator are characterized over the full dynamic range of the attenuator. The test was performed on 6 April 2026 in the lab.

2. METHODS

2.1. *Signal Chain*

The input signal is generated by a Micronetics NOD-5108 noise diode with a nominal output PSD of −75 dBm/Hz. The signal passes through a fixed −11 dB attenuator and a low-pass filter, and is then split with a Mini-Circuits ZFSC-2-2500 power splitter into the two SNAP ADC inputs N0 and E2. The noise diode has a built-in step attenuator that can be varied from 0 to 110 dB. With the 225 MHz analog bandwidth set by the LPF, the total integrated input power at 0 dB attenuation is +8.5 dBm.

2.2. *Test Setup*

Data are acquired with the EIGSEP F-engine firmware `eigsep_fengine_1g_v2_3_2024-07-08_1858.fpg` running on a SNAP board sampled at 500 MHz with 1024 frequency channels. The correlator is configured with `adc_gain = 4`, `fft_shift = 0x015F`, `corr_scalar = 512`, and `corr_acc_len =` 2^{28}, giving an accumulation cadence of roughly one spectrum per second. The relevant software versions are `eigsep_observing f1d286ba0`, `eigsep_corr d8410b767`, and `casperfpga ea9701f3`. The board is programmed and initialized once with `fpga_init.py -Pasf`.

2.3. *Measurement Procedure*

For each attenuator setting we record two quantities:

1. The RMS of the raw ADC samples, computed from 20 ADC snapshots acquired with `adc_snapshot.py --nsamples 20`, sweeping the attenuator from 0 to 110 dB.
2. The total integrated autocorrelation power on inputs N0 and E2, obtained from a ∼10 s integration per attenuator step using `corr_linearity.py`.

3. RESULTS

Figure 1 shows the measured ADC RMS (left) and the total autocorrelation power (right) as a function of the input power at the splitter output. Both inputs N0 and E2 track each other to within a few percent across the full sweep, as expected for a passive split of the same noise source.

The ADC RMS is linear in input power from approximately −32 dBm up to +8 dBm. Below −32 dBm the noise diode contribution falls below the intrinsic ADC noise floor (RMS ≈ 0.48 counts) and the curve flattens. The autocorrelation power is linear over a slightly narrower range, from approximately −27 dBm to +8 dBm, with a comparable floor set by the digital noise of the F-engine. No compression is observed at the high end of the sweep, indicating that the SNAP front end and digitizer remain linear up to at least +8 dBm of integrated input power for this signal bandwidth.

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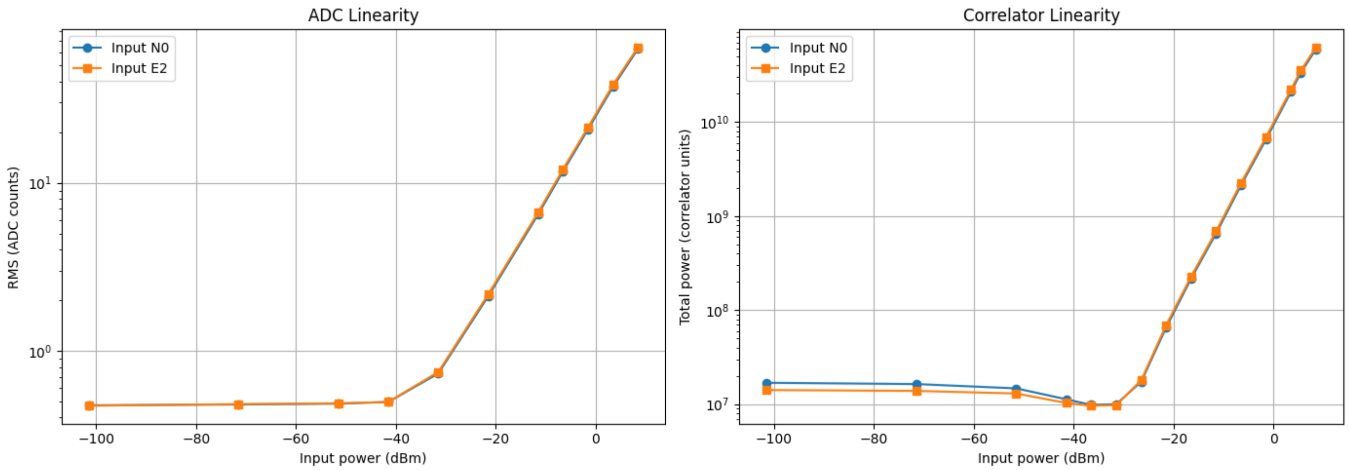

Figure 1. SNAP linearity measurements. *Left:* RMS of the raw ADC samples versus input power for inputs N0 and E2. *Right:* Total integrated autocorrelation power in correlator units versus input power for the same two inputs. The ADC RMS is linear from ~ -32 dBm to $+8$ dBm, and the autocorrelation power is linear from ~ -27 dBm to $+8$ dBm.